



ABODE SOFTWARE • PROJECT TECHNICAL PRESENTATION

DevOps Lifecycle Implementation & Pipeline Automation

Automated configuration management, git branching model, Docker containerization, and multi-stage Jenkins CI/CD pipelines.

ROLE

Sr. DevOps Engineer

TARGET REPOSITORY

github.com/hshar/website.git

Core Operational Requirements



Configuration Management

Automated server setup using Ansible host playbooks.



Git Branch Workflow

Isolated branch execution rules for Master & Develop.



Docker Packaging

Pre-built base container hshar/webapp deployed to /var/www/html.



Jenkins Orchestration

Automated Build, Test, and Prod deployment stages.

DevOps Lifecycle Architecture

1. Source Control Management

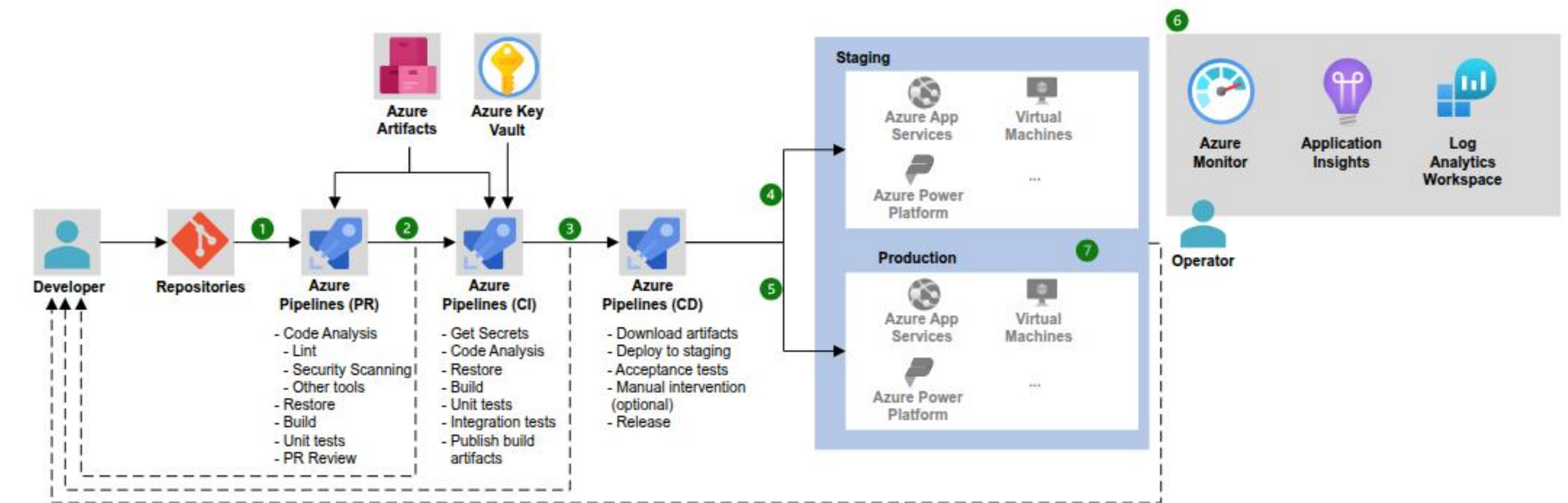
GitHub repository webhook triggering automated CodeBuild jobs.

2. CI/CD Orchestration

Jenkins master running multi-job pipeline executions.

3. Container Runtime Environment

Docker image mapping application root to /var/www/html.



Task 1: Configuration Management

Ansible / Puppet

Automated Node Provisioning

- ✓ **Automated Runtime Setup:** Continuous installation of Docker engine and Git tools.
- ✓ **Declarative Inventory Management:** Standardized server profiles across target nodes.
- ✓ **Zero Drift Execution:** Idempotent playbooks preventing manual server drift.

Ansible Provisioning Snippet

```
- name: Install Runtime Stack
hosts: all_nodes
tasks:
  - apt: name=docker.io state=present
  - apt: name=git state=present
  - apt: name=jenkins state=present
```

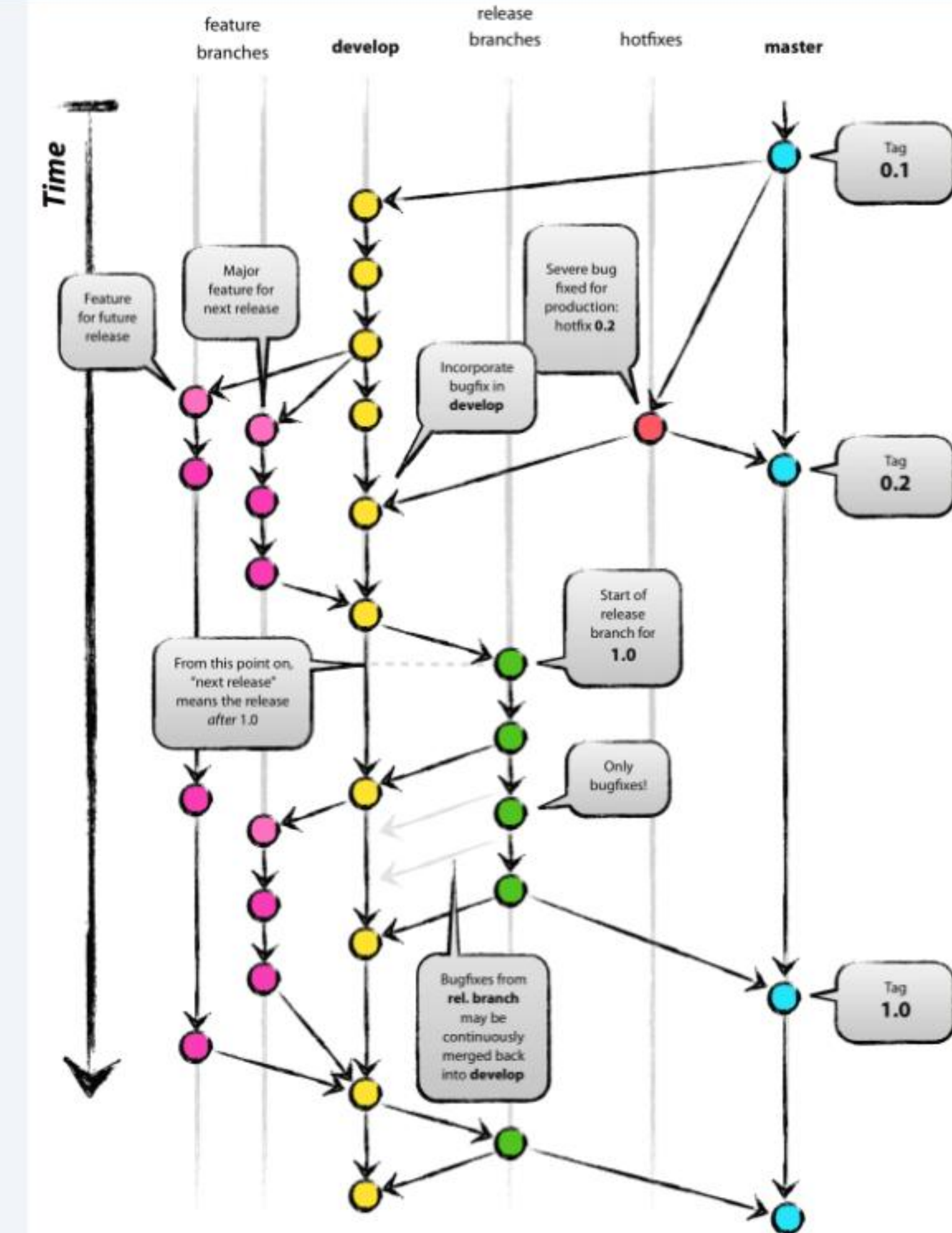
Task 2 & 3: Git Branching & Webhook Triggers

Master Branch Workflow

Triggers CodeBuild -> Executes Tests -> Automated Deployment to Production.

Develop Branch Workflow

Triggers CodeBuild -> Executes Tests -> Halts (No Production Push).



Branch Execution Matrix

master

Production Path

- ✓ CodeBuild Triggered on Commit
- ✓ Automated Test Suite Execution
- ✓ **Push to Production Server**

develop

Staging Path

- ✓ CodeBuild Triggered on Commit
- ✓ Automated Test Suite Execution
- ✗ **Halt Deployment (No Prod Push)**

Task 4: Docker Container Strategy

BASE IMAGE SPECIFICATION

hshar/webapp

Pre-configured Apache web server container image.

TARGET DIRECTORY PATH

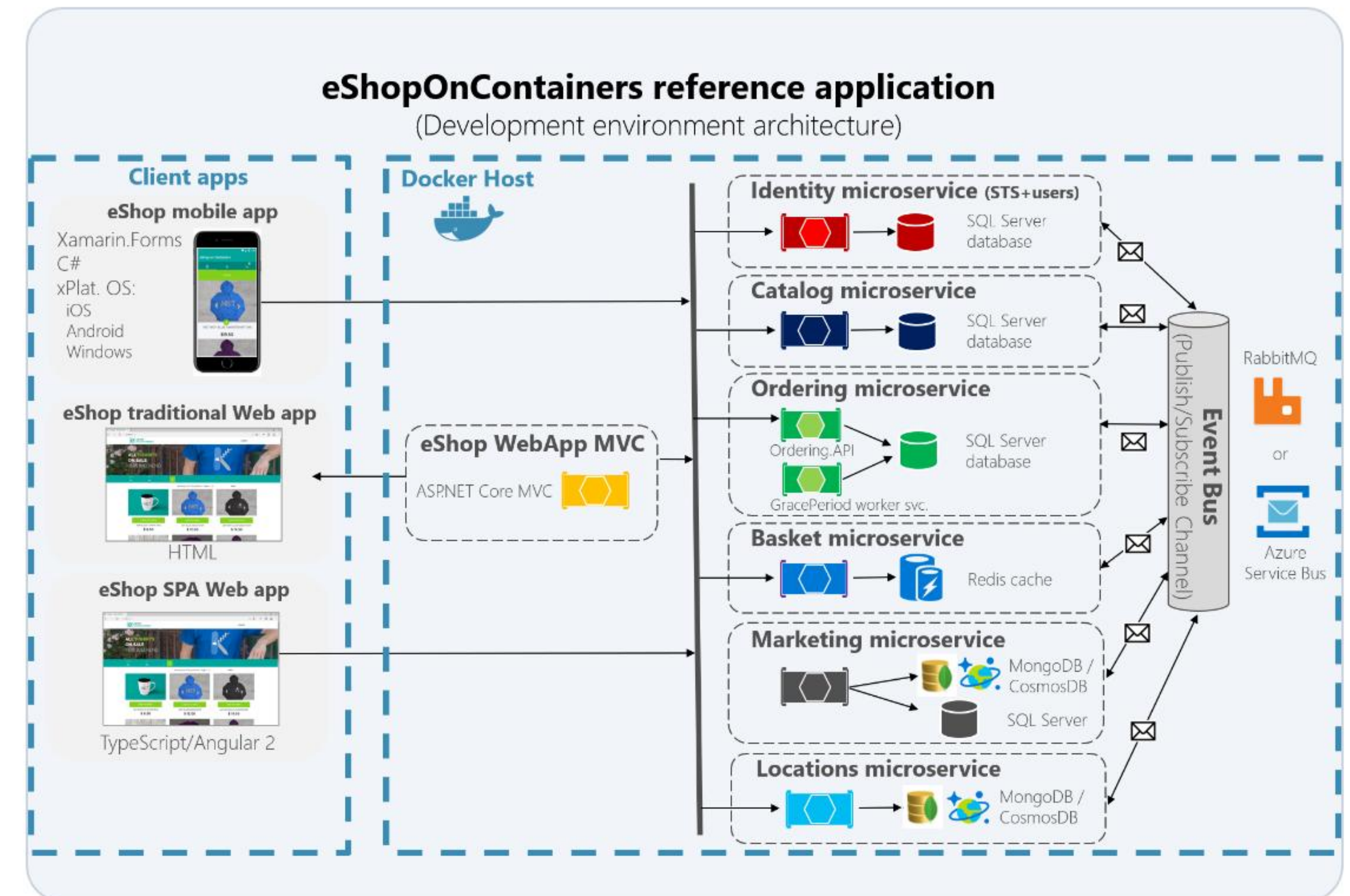
/var/www/html

Application code mounted into container web root.

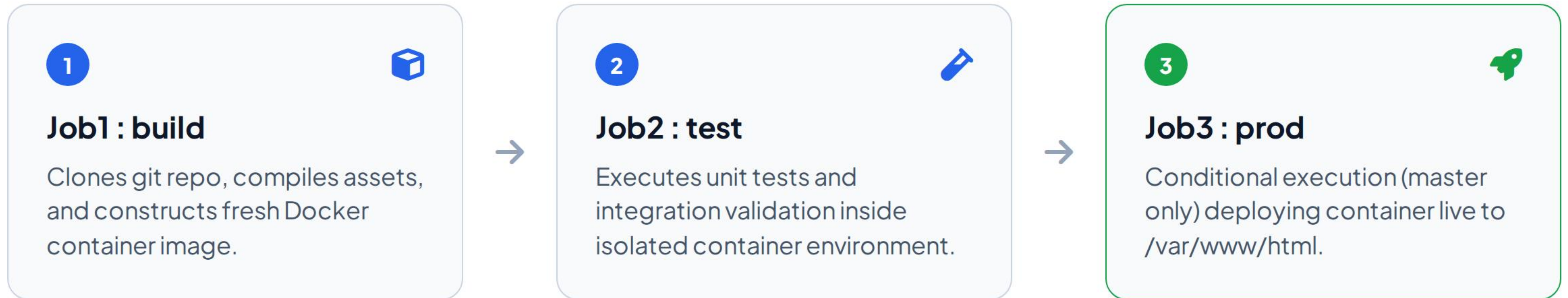
TRIGGER AUTOMATION

GitHub Push Event

Automated Docker build triggered on git push events.



Task 5: Jenkins Multi-Job Pipeline



Implementation Benefits for Abode Software

100%

Automated CI/CD

Zero manual interventions from
commit to deploy.

0 Drift

Consistent Envs

Identical runtime setup using
Docker containers.

< 5m

Build Cycle

Rapid feedback loop for
production releases.



DevOps Implementation Ready for Deployment

All five specification requirements mapped, containerized, and configured into an automated multi-job Jenkins pipeline.

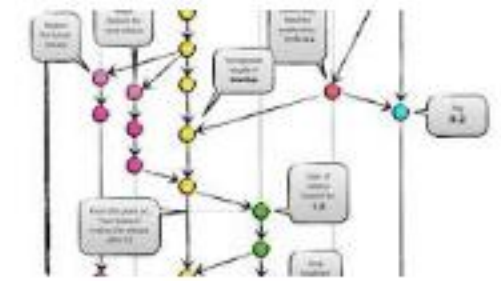
Abode Software DevOps Transition Plan

Image Sources



<https://learn.microsoft.com/en-us/azure/devops/pipelines/architectures/media/azure-devops-ci-cd-architecture.svg?view=azure-devops>

Source: learn.microsoft.com



<https://nvie.com/img/git-model@2x.png>

Source: nvie.com



https://devblogs.microsoft.com/dotnet/wp-content/uploads/sites/10/2017/08/eShopOnContainers_Architecture_Diagram.png

Source: devblogs.microsoft.com